

Maru Metall — Business Model Canvas

SteelQuote AI · 28DIGITAL Business Challenge 2026

PRODUCT NAME

SteelQuote AI

An AI-supported cost estimation tool that interprets mixed-quality project documentation (IFC models, PDF drawings, material lists, specifications) and generates a structured, traceable cost quotation that an experienced estimator can review, adjust, and send — in a fraction of the time required by the manual process.

CHALLENGE CONTEXT

Maru Metall receives 500+ price inquiries per year. Each requires manual interpretation of heterogeneous inputs (IFC, BOM, PDF, DWG, email) and expert translation into a detailed cost estimate covering: material, fabrication labour (plasma cutting, sawing, drilling, assembling, welding), surface treatment (paint systems C2M–C4H, HDG), packaging and transport. The process is slow, expert-dependent, and not scalable.

SteelQuote AI addresses this directly, addressing all four challenge tasks: 1. Input interpretation 2. Quantity and effort extraction 3. Risk and assumption flagging 4. Traceable output generation

1. CUSTOMER SEGMENTS

Primary user

Maru Metall's cost estimation team — specifically senior cost engineers and sales managers who currently perform manual quotation work. The product is internal-facing at launch.

Secondary beneficiaries

- **Sales and business development team** at Maru: faster quotes = more bids = more revenue opportunity.
- **Project managers**: faster go/no-go decisions on inquiry intake.
- **Board / management**: capacity to scale bid volume without proportional headcount increase.

Long-term expansion

Other Nordic steel fabricators facing identical operational challenges (the model is replicable across the industry).

2. VALUE PROPOSITION

For cost engineers: - Interpret project documentation in minutes rather than hours. - Receive a structured draft estimate with every assumption made explicit. - Focus review time on complex decisions and client-specific adjustments — not on mechanical quantity extraction.

For sales management: - Quote more projects in the same time window. - Reduce the dependency on a small number of senior estimators. - Improve bid-to-win ratio by responding faster to inquiries.

For the business: - Recover hundreds of engineering hours per year. - Scale bid capacity without proportional cost increase. - Improve margin control through more consistent, norm-based cost logic.

The core promise: SteelQuote AI does not replace the experienced estimator. It eliminates the mechanical work so the estimator can focus on judgment, client relationship, and margin protection.

3. CHANNELS

Phase 1 — Internal deployment (PoC): - Direct deployment within Maru's estimation team. - Used on live incoming inquiries in parallel with the manual process. - Accuracy measured against actual quotes produced.

Phase 2 — Internal roll-out: - Full integration into Maru's quotation workflow. - Connected to existing ERP / project management systems. - Trained on historical Maru quotes to improve accuracy over time.

Phase 3 — External licensing (optional): - License the tool to other steel fabricators in non-competing geographies. - SaaS model: per-inquiry fee or annual subscription.

4. CUSTOMER RELATIONSHIPS

Stage	Mode	Mechanism
PoC	Human-in-the-loop	Estimator reviews every AI output; feedback loop trains the model
Production use	AI-assisted	AI drafts, estimator reviews and adjusts exceptions
Mature use	AI-primary	AI handles standard inquiries autonomously; human reviews complex cases
External licensing	SaaS support	Onboarding, accuracy calibration, ongoing updates

The relationship model is designed to build trust gradually: the tool earns autonomy as its accuracy is demonstrated, not assumed.

5. REVENUE STREAMS

Internal value (Phase 1–2): - Cost savings from recovered engineering hours. - Revenue increase from higher bid volume. - Margin improvement from more consistent pricing logic.

External value (Phase 3, optional):

Stream	Model	Estimate
SaaS licensing to other fabricators	Annual subscription	€15k–€60k/year per client
Per-inquiry processing fee	Usage-based	€50–€200 per quote generated
Implementation and calibration services	Professional services	€10k–€30k per client onboarding

6. KEY RESOURCES

Resource	Why it matters
Historical Maru quotations	Training data — the ground truth for what a correct estimate looks like
Cost estimation norm database	The production logic (labour norms, surface treatment rates, transport rates) from the principles workbook
IFC / BOM parsing capability	The technical core — ability to read and interpret structural data formats
LLM / AI inference capability	The interpretation layer for unstructured inputs (PDF drawings, emails, specs)
Experienced estimator validation	Human feedback loop that improves model accuracy over time
EN 1090-2 domain knowledge	The regulatory standard that governs all fabrication quality and scope assumptions

7. KEY ACTIVITIES

Activity	Description
Document interpretation	Parse IFC, BOM, PDF, DWG, email to extract fabrication scope
Quantity extraction	Identify element types, profiles, steel grades, weights, lengths
Labour estimation	Apply norm-based production logic: cutting, sawing, drilling, assembling, welding
Surface treatment pricing	Map corrosion class and paint system to m2-based cost model

Activity	Description
Risk and assumption flagging	Identify and document every information gap, inconsistency, or exclusion
Output generation	Produce structured, line-item cost estimate in Maru's standard format
Accuracy measurement	Compare AI output to actual quote; feed delta back into training

8. KEY PARTNERS

Partner	Role
AI / LLM provider	Core language model for document interpretation
IFC / BIM software ecosystem	Data format standards and parsing libraries
Maru's existing ERP system	Integration for historical data and output delivery
EN 1090-2 standard body	Regulatory framework embedded in the estimation logic
External steel fabricators (Phase 3)	Future licensing clients

9. COST STRUCTURE

Cost category	Description
AI infrastructure	LLM API usage, inference compute, vector database for document storage
Development	Software engineering for parsing, estimation logic, UI, ERP integration
Training data curation	Cleaning and structuring historical Maru quotes for model training
Validation and calibration	Estimator time spent reviewing PoC outputs and providing feedback
Ongoing maintenance	Model updates, new input format support, accuracy monitoring

PoC cost (minimal viable version): The PoC does not require a full software platform. It can run as a structured pipeline: document parser (open-source IFC/PDF tools) + LLM prompt chain + norm-based calculation engine in Python/Excel. Initial build is measured in days, not months.

CANVAS SUMMARY

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITION	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
AI/LLM provider	Document interpretation	Estimates in minutes not hours	Human-in-loop	Marine engineering
IFC/BIM ecosystem	Quantity extraction	Every assumption explicit	AI-assisted	Salvage
Maru ERP system	Labour estimation	Scalable bid capacity	AI-primary (mature)	Marine
EN 1090-2 bodies	Surface treatment price	Margin consistency	SaaS support (external)	Future fabricators
External fabricators	Risk/assumption flag	Estimator freed for judgment		
KEY RESOURCES			CHANNELS	
Historical quotes	COST STRUCTURE		Internal deploy	
Norm database		REVENUE STREAMS	ERP integration	
IFC/BOM parsers	AI infrastructure	Hour recovery (internal)	External SaaS licensing	
LLM capability	Development	Higher bid volume		
Estimator feedback	Data curation	SaaS licensing (ext.)		
EN 1090-2 knowledge	Validation/calibration	Per-inquiry fee		
	Maintenance			

EVALUATION AGAINST JUDGING CRITERIA

Criterion (25%)	Evidence
Problem & Value Proposition	Pain is precise and documented: 500+ inquiries/year, hundreds of hours each, expert-dependent. Value is concrete: hours recovered, bids scaled.
Business Model & Viability	All 9 BMC blocks complete. Revenue from internal efficiency gains (Phase 1–2) and optional external SaaS (Phase 3). Cost of PoC is minimal.
Go-to-Market & Feasibility	Internal deployment first eliminates market risk. PoC uses real Maru data (Example 4 BOM). No customer acquisition required for Phase 1.
Pitch Quality & Q&A	Story maps directly to the 4 challenge tasks. Numbers grounded in real project data. PoC is concrete and demonstrable.